ERTMS/ATO
Test Specification
REF : SUBSET-151 ISSUE : 1.0.0 DATE : 09-12-24

Company	Technical Approval	Management approval
ALSTOM		
AZD		
CAF		
HITACHI RAIL		
MERMEC		
SIEMENS		

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1. MODIFICATION HISTORY

Issue Number Date	Section Number	Modification / Description	Author
0.0.1 15-10-22	1..5	Creation	UNISIG ATO WP
0.0.2 21-10-22	6.3, 7	Changed title and subset number Added test specifications	UNISIG ATO WP
0.0.3 03-11-22	All	Update after first round of ATO WP comments	UNISIG ATO WP
0.0.4 04-11-22	All	Update during ATO WP joint meeting	UNISIG ATO WP
0.0.5 09-11-22	All	Added traceability matrix	UNISIG ATO WP
0.0.6 10-11-23	§5.1	Refining the scope of the document according to the agreement from the joint meeting of the 5/10/23 involving UNISIG, EUG and ERA	UNISIG ATO WP
	§5.2	Update of the reference document [REF12] referencing the SUBSET-110	UNISIG ATO WP
	§5.2	Adding a reference referring to the SUBSET-111	UNISIG ATO WP
	§7.11	Sequence diagram related to the scenario 11 has been updated	UNISIG ATO WP
	§7.12	Sequence diagram related to the scenario 11 has been updated	UNISIG ATO WP
	§7.13	Sequence diagram related to the scenario 11 has been updated	UNISIG ATO WP
0.0.7 18-12-23	§8.1.1.1	Updating the Traceability Matrix with the list of	UNISIG ATO WP

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		requirements covered by the Test Cases	
0.0.8 12-02-24	§§ 5.1.1.9; 5.1.1.13; 5.1.1.13.3; 5.1.1.13.4; 5.1.1.13.5; Table 1 REF[12] and REF[13]	Reworded according to ERA comments from the 18/12/2023	UNISIG ATO WP
0.0.9 22-04-24	§ 5.1.1.18 d) § 8.2	Removed because the traceability regarding to SS-126 not needed.	EUG UNISIG ATO-WP
	§ 6.1	Clarification given on test cases regarding to Test Cases. A new template is used for test cases specification	
	§ 6.3	The list of test cases has been updated	
	§ 7	Updated with a new template according to SUBSET-125 v1.0.0	
	§ 8	The traceability matrix has been updated	
0.0.10	All	Upgraded according to the agreed comment from the consolidated Review Sheet dedicated to the version 0.0.9.	
0.0.11		Upgraded according to the agreed comment from the consolidated Review Sheet dedicated to the version 0.0.10.	
0.1.0 08-08-24	Template § 8	Change of template. Replacement of non readable table with its source Excel file (Traceability Matrix).	UNISIG
1.0.0 09-12-24		Baseline 4 – 1st Release version	

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5. INTRODUCTION

5.1 Scope and purpose of the document

5.1.1.1 ERTMS/ATO is a part of CCS subsystem for which responsibilities and functions have been apportioned between the ETCS-OB, ATO-OB and ATO-TS equipment.

5.1.1.2 ATO provides a set of non-safety functions related to speed control, accurate stopping, doors opening/closure, and other functions traditionally assigned to a driver, while ETCS ensures the safety of operation in cooperation with other external systems (i.e. door management systems).

5.1.1.3 The system requirements for an interoperable GoA2 application of ERTMS/ATO have been specified in [Ref 3] and [Ref 4].

5.1.1.4 Note: in GoA2 the train is driven automatically, stopping is automated but a driver in the cab is required to start automatic driving of the train and to check the clearance of the track ahead. The driver can operate the doors (although this can also be done automatically), can take over in emergency or degraded situations and can carry out other manual functions.

5.1.1.5 Since ERTMS/ATO equipped trains shall be able to operate over any ERTMS/ATO equipped track, the interface between ERTMS/ATO on-board and trackside must be technically interoperable.

5.1.1.6 For interoperable interfaces between trackside and on-board the CCS TSI requires that their implementation (functions, protocols, electrical and physical aspects) complies with the mandatory specifications i.e. messages, commands, data etc. communicated between them must be understood by the receiver as intended by the sender.

5.1.1.7 ERTMS/ATO track to train interface has been specified in [Ref 5] (application layer) and [Ref 6] (transport layer).

5.1.1.8 The verification of ERTMS/ATO track to train interface is not within the scope of this testing. Therefore, the requirements of the test environment for this interface are out of the scope of this document.

5.1.1.9 Although no specific tests for the communication layers ([Ref 6]) have been provided, the scenarios defined for the application layer rely on the nominal behaviour of the lower communication layers (connection establishment, transmission of application data, etc.) that are therefore generally tested through this test plan.

5.1.1.10 The scope of this document is to define a Test Specification for the ERTMS/ATO to ensure the interoperability between ATO-TS and ATO-OB parts according to system requirements defined in [Ref 3], [Ref 4] and [Ref 10].

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5.1.1.11 The objective of the document is to test and verify the interoperability between ATO-OB and ATO-TS:

- If an ATO-TS passes all Test Cases, the ATO-TS is considered track/train interoperable according to [Ref 4].
- If an ATO-OB passes all Test Cases, the ATO OB is considered track train interoperable according to [Ref 4].

5.1.1.12 The tests are focussing only on the ATO-OB and ATO-TS interoperable interface. Other subsystems (and their related interfaces) are not tested as such. They may be used or simulated to support the ATO related tests.

5.1.1.13 Even if the requirements from [Ref 4] related to the following interfaces are identified within the attached excel file supporting the traceability matrix, these requirements are not covered by this document:

a) ETCS-OB / ATO-OB Interface ([Ref 7], [Ref 8]);

b) ATO-OB / Rolling Stock Interface ([Ref 9]).

5.1.1.14 It is recommended to perform tests involving an ATO-TS and ATO-OB configuration representing the final set of application specific segment profiles and train characteristics.

5.1.1.15 The **purpose** of this document is to specify:

a) the methodology followed for the definition of the minimum set of Test Cases needed to ensure track to train interoperability;

b) the specification for the Test Cases;

c) the traceability matrix between the Test Cases and the interoperable requirements of [Ref 4].

5.1.1.16 This test specification is applicable to ATO-OB and ATO-TS compliant with ATO system version defined in the [Ref 4].

5.2 Reference documents

Ref. N°	Title	Reference
[Ref 1]	ERTMS/ATO Glossary	13E154
[Ref 2]	ERTMS/ATO Operational Scenarios	13E151
[Ref 3]	ERTMS/ETCS System Requirements Specification	SUBSET-026
[Ref 4]	ERTMS/ATO System Requirements Specification	SUBSET-125
[Ref 5]	ATO-OB / ATO-TS FFFIS Application Layer	SUBSET-126

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Ref. N°	Title	Reference
[Ref 6]	ATO-OB / ATO-TS Interface Specification - Transport Layers	SUBSET-148
[Ref 7]	ETCS-OB / ATO-OB FFFIS Application Layer	SUBSET-130
[Ref 8]	ATO-OB Interface Specification Communication Layers for On-board Communication	SUBSET-143
[Ref 9]	ATO-OB / Rolling Stock FFFIS Application Layer	SUBSET-139
[Ref 10]	ETCS Driver Machine Interface	ERA_ERTMS_015560
[Ref 11]	Glossary of Terms and Abbreviations	SUBSET-023

Table 1 - Reference Documents

5.3 Abbreviations

5.3.1.1 For ATO related abbreviations see ERTMS/ATO Glossary [Ref 1].

5.3.1.2 For ETCS related abbreviations see SUBSET-023 [Ref 11].

5.4 Definitions

5.4.1.1 For ATO related definitions see ERTMS/ATO Glossary [Ref 1].

5.4.1.2 For ETCS related definitions see SUBSET-023 [Ref 11].

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5.5 ERTMS/ATO Reference Architecture

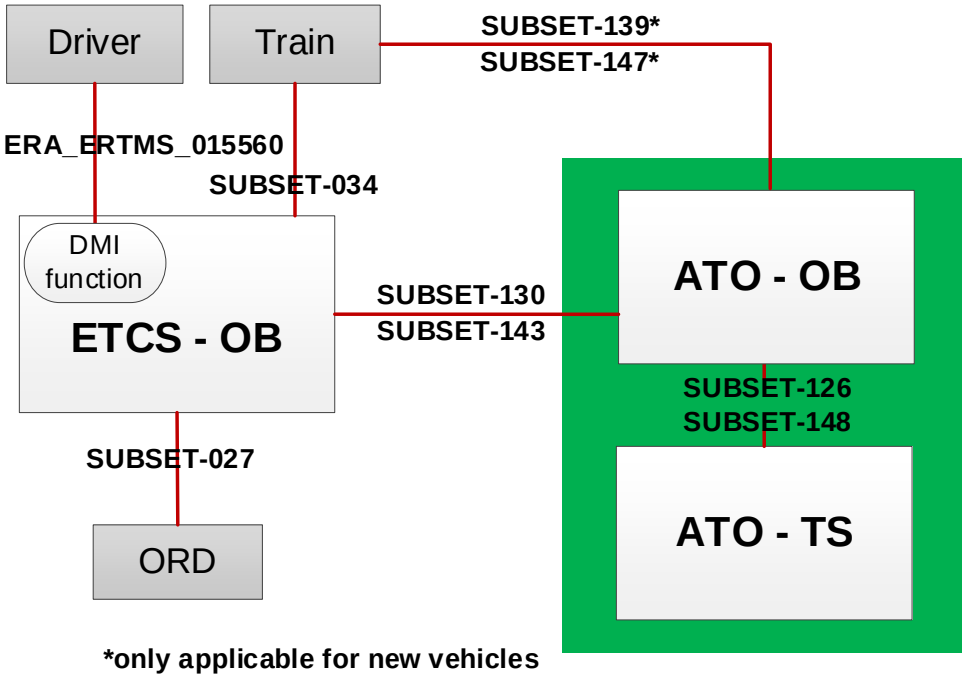


Figure 1 - ERTMS/ATO reference architecture

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6. TEST PLAN

6.1 Methodology

6.1.1 Test Cases

6.1.1.1 Any GoA2 function specified by EEIG ERTMS Users Group (e.g. automatic turnback) but not yet included in the current version of [Ref 4] has been excluded from this document.

6.1.2 Device under test

6.1.2.1 When performing the Test Case, the DUT (Device Under Test) may be the ERTMS/ATO on-board, the ERTMS/ATO trackside or both, depending on which constituent shall prove its interoperability.

6.1.3 Test Cases presentation

6.1.3.1 For each Test Case, a basic description provides the following information:

a) Scenario Description: the scope and the operational context of the Test Case.

b) Test Case steps definition with the related condition for being successfully passed.

c) Notes: additional information to facilitate the test execution.

6.1.3.2 The actors defined for the test steps are:

a) ERTMS/ATO on-board

b) ERTMS/ATO trackside

c) The operational staff (e.g. driver)

d) External systems (e.g. the train, the Traffic Management System)

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6.1.3.3 The following table gives an example of test Case template:

ID	Date	Location / Line
1	<YYYY-MM-DD>	<Line>
Title		
Communication session establishment		
Description		
This describes the operational activities required to establish a communication session when in NA State, either after energising the ATO onboard or after a change of TRN.		
Sequences of the Test Case		
Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB detects that ETCS-OB sends validated Train Data			
2	ATO-OB: ATO-OB sends a HSReq to ATO-TS using the pre-configured Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
3	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
4	ATO-OB: Communication session established (condition to end scenario)			
Notes		-		

6.1.4 Pre and post conditions

6.1.4.1 Each Test Case shall be defined with preconditions and postconditions to fulfil.

6.1.4.2 The Figure 2 shows the preconditions and postconditions applicable for all the Test Cases.

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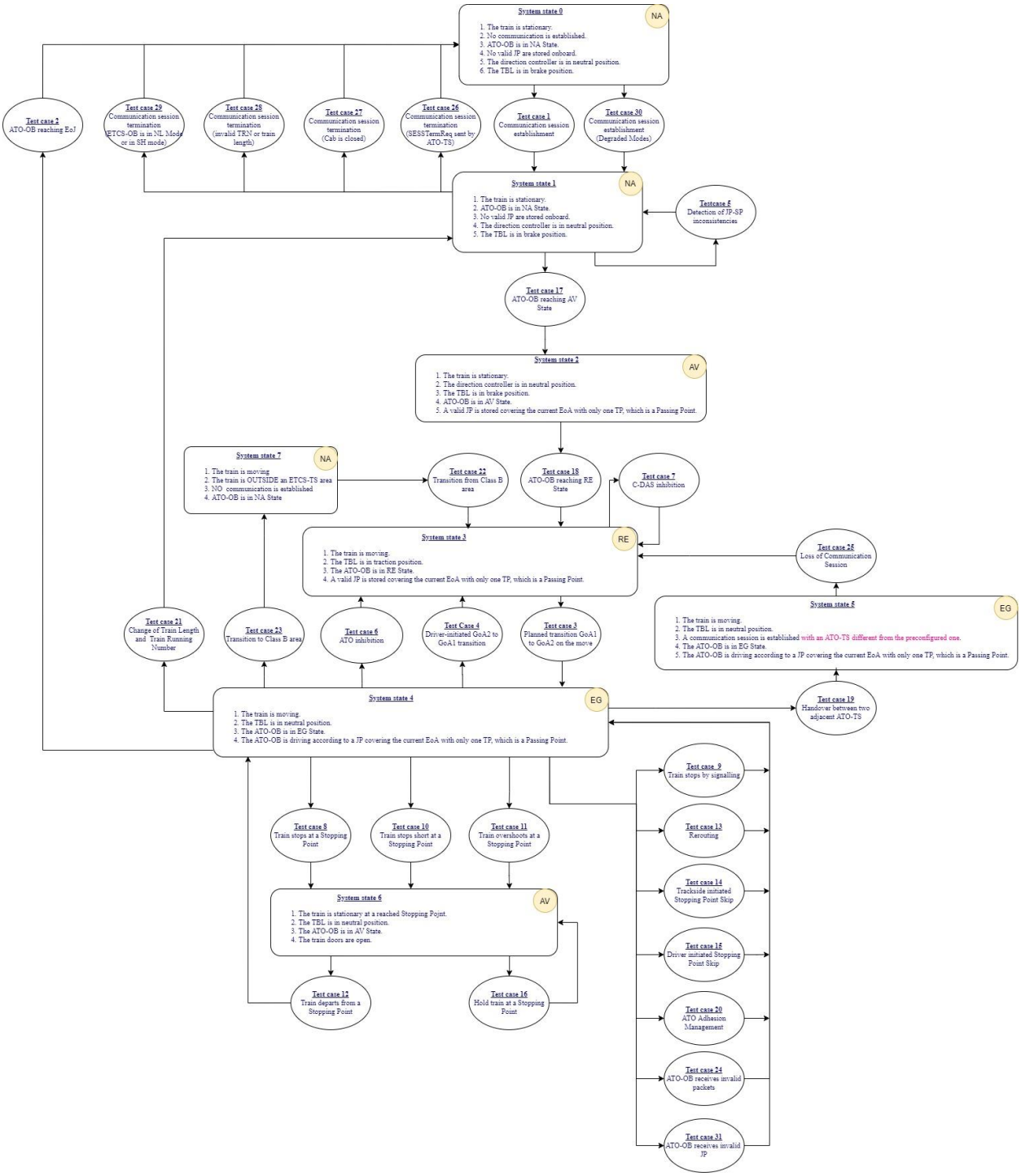


Figure 2 – Specific pre and postconditions associated with Test Cases

6.1.4.3 The Figure 3 specifies generic conditions which are applicable to all the Test Cases as pre and postconditions except it is specified otherwise.

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System state generic

(always applicable except specified otherwise)

1. All vehicle and OB systems are fully operational.
2. The CAB is opened.
3. The ETCS-OB has a valid position.
4. The train is not located in ATO Inhibition zone, DAS inhibition zone, neutral/powerless section.
5. The ETCS-OB SoM is completed and the ETCS-OB is in FS Mode.
6. Cyclic reporting is configured at the maximum time interval (7.7.1.3.cc of [Ref. 4]).
7. The Operational Conditions "No emergency brake applied" is fulfilled.
8. The train applicable conditions for operating ATO are fulfilled.
9. There is no passenger on-board (passengers train)/no more cargo loading/unloading will take place (cargo train).
10. A communication session established with the preconfigured ATO-TS.
11. No Segment Profile is stored in the ATO-OB at the begining of test scenario

Figure 3 – Generic conditions

6.1.4.4 Finally, the Figure 4 specifies the expected generic behaviour of ATO-OB which is not specified in the Test Cases.

Generic ATO-OB behaviour

(always applicable except specified otherwise)

1. When approaching the end of the current JP, the train shall request JP extension (via a JPReq) in background.
2. when needed SP are missing supporting a test case, ATO-OB shall request them and acquire them in background.
3. The ETCS-OB MA are automatically extended when the train is moving except for the scenario related to the train stop due to signalling

Figure 4 – Expected generic ATO-OB behaviour

6.1.4.5 Test Scenarios can be built according to different trackside configurations without affecting the test result.

6.1.4.6 The Test Scenarios will be based on:

- The trackside configuration;
- The possible successions of Test Cases as they appear on the Figure 2.

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6.2 ERTMS/ATO GoA2 tested functions

6.2.1 Track to train interoperable functions

6.2.1.1 The following ATO functions will be stimulated during the execution of tests defined in this specification:

1) Time table management
2) Supervised Speed Envelope Management
3) Automatic Train Stopping Management
4) ATO Traction / Brake Control
5) Timing Point Management
6) Train Door Operation
7) Dwell time Management
8) Skip Stopping Point
9) Train Hold at Stopping Point
10) Low Adhesion Management
11) Time Management
12) Reporting Management
13) Routing Errors
14) Driver Advisory System (DAS)
15) Perform ATO-OB self tests
16) ATO-OB Data acquisition (ETCS data)
17) Driver Machine Interface: Information displayed to the driver

6.3 List of Test Cases

6.3.1.1 The tests cases identified in this test plan for the verification of the ERTMS/ATO interoperability are:

1) Communication session establishment
2) ATO-OB reaching EoJ
3) Planned GoA1 to GoA2 transition on the move
4) Driver-initiated GoA2 to GoA1 transition
5) Detection of JP-SP inconsistencies
6) ATO inhibition
7) C-DAS inhibition
8) Train stops at a Stopping Point
9) Train stopped by signalling
10) Train stops short at a Stopping Point

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11) Train stops overshoots Stopping Point
12) Train departs from a Stopping Point
13) Rerouting
14) Trackside initiated “Stopping Point Skip”
15) Driver initiated “Stopping Point Skip”
16) Hold train at a Stopping Point
17) ATO-OB reaching AV State
18) ATO-OB reaching RE State
19) Handover between two adjacent ATO-TS
20) ATO Adhesion Management
21) Change of Train Running Number
22) Transition from Class B area
23) Transition to Class B area
24) Invalid packet received by ATO-OB
25) Loss of communication Session
26) Communication session termination (SESSTermReq sent by ATO-TS)
27) Communication session termination (Cab is closed)
28) Communication session termination (invalid TRN or train length)
29) Communication session termination (ETCS-OB is in NL Mode or in SH mode)
30) Communication session establishment (Degraded Modes)
31) ATO-OB receives invalid JP

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7. TEST CASES SPECIFICATION

7.1 Communication session establishment

ID	Date	Location / Line
1	<YYYY-MM-DD>	<Line>
Title		
Communication session establishment		
Description		
This describes the operational activities required to establish a communication session when in NA State, either after energising the ATO onboard or after a change of TRN.		
Sequences of the Test Case		
Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB detects that ETCS-OB sends validated Train Data			
2	ATO-OB: ATO-OB sends a HSReq to ATO-TS using the pre-configured Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
3	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
4	ATO-OB: Communication session established (condition to end Test Case)			
Notes		-		

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7.2 ATO-OB reaching EoJ

ID	Date	Location / Line
2	<YYYY-MM-DD>	<Line>
Title		
ATO-OB reaching EoJ		
Description		
This describes the operational activities required at the end of an ATO train's journey.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with only one additional Timing Point, which is a Stopping Point with the attribute EoJ	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB approaches the Stopping Point which has the attribute EoJ and considers the Stopping Point as reached		Pass / Fail	
5	ATO-OB: ATO-OB detects that Communication session transition [3] is fulfilled and sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition 1 to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
6	ATO-OB: ATO-OB detects that the Operational Conditions are lost			
7	ATO-OB: ATO-OB transitions to DE State			
8	ATO-OB: before 5 seconds have passed since reaching the Stopping Point, the ATO-OB detects that the train has come to a stop	Accurate stop displayed on ETCS-OB DMI.	Pass / Fail	
10	ATO-OB: ATO-OB transitions to NA State (condition 2 to end Test Case)	End of Test Case.		
Notes		- The de-energising of the ATO-OB after the termination of the communication session is excluded from this Test Case, as it is strictly related to the Rolling Stock capabilities		

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7.3 Planned GoA1 to GoA2 transition on the move

ID	Date	Location / Line
3	<YYYY-MM-DD>	<Line>
Title		
Planned GoA1 to GoA2 transition on the move		
Description		
This describes the operational activities required for a train to transition into GoA2 on the move.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	Operational Staff: The driver selects “ATO Engage”			
2	External Systems: ETCS-OB will transition to AD Mode and report to ATO-OB that “ATO Engage” has been selected (by increasing the related counter in packet 7 in SUBSET-130)			
3	ATO-OB: ATO-OB detects that “ATO Engage” has been selected			
4	ATO-OB: ATO-OB transitions to EG State			
5	Operational Staff: The driver moves the TBL to neutral position (condition 1 to end Test Case)			
6	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in EG State (condition 2 to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in EG State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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7.4 Driver-initiated GoA2 to GoA1 transition

ID	Date	Location / Line
4	<YYYY-MM-DD>	<Line>
Title		
Driver-initiated GoA2 to GoA1 transition		
Description		
This describes the operational activities required when leaving GoA2 after a brake request from the driver. The ATO-OB shall go to GoA1 being in RE State.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	Operational Staff: The driver moves the TBL to brake position			
2	ATO-OB: ATO-OB detects that the TBL is in brake position (and that the Engagement conditions are lost)			
3	ATO-OB: ATO-OB transitions to AV State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in AV State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	Operational Staff: The driver moves the TBL to traction position			
5	ATO-OB: ATO-OB detects that the Engagement conditions are fulfilled			
6	ATO-OB: ATO-OB transitions to RE State			
7	ATO-OB: ATO-OB will request the ETCS-OB to display DAS Information.	“Target Advice Speed” or “Coasting advice”, and “Distance to next advice change” are displayed on ETCS-OB DMI.	Pass / Fail	
8	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in RE State (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in RE State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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7.5 Detection of JP-SP inconsistencies

ID	Date	Location / Line
5	<YYYY-MM-DD>	<Line>

Title

Detection of JP-SP inconsistencies

Description

This describes the test sequences supporting the different type of inconsistencies which could occur:

- TP included in the JP is not found in the corresponding SP;
- The ATO-TS sends an SP with the status “invalid”;
- Time references of TPs listed in the JP are equal, or not in ascending order;
- The SP does not provide the location values in increasing order for any profile data type;
- Any location data given in an SP is longer than the length of the SP;
- Any length data given in the SP is zero;
- The SP version of an SP received is not the one included in the corresponding JP.

Test Cases

Sequences of the Test Case

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS with an undefined SP as a reference	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: Upon reception of the JP, the ATO-OB will start to monitor the triggering conditions for sending STRs			
5	ATO-OB: ATO-OB detects that one TP mentioned in the JP is not found in the corresponding SP		Pass / Fail	
6	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
7	ATO-TS: ATO-TS sends a JP Overwrite	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
8	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
9	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP that are not stored in ATO-OB.	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
10	ATO-TS: ATO-TS sends the requested SPs, where one SP has the status “invalid”	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
11	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
12	ATO-TS: ATO-TS sends a JP Overwrite where the time references of two TPs are not in strictly ascending order	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
13	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
14	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
15	ATO-TS: ATO-TS sends a JP Overwrite	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
16	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
17	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP that are not stored in ATO-OB.	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
18	ATO-TS: ATO-TS sends the requested SPs, where one SP does not provide the location values in increasing order for one profile data type	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
19	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
20	ATO-TS: ATO-TS sends a JP Overwrite	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
21	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	

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22	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP that are not stored in ATO-OB.	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
23	ATO-TS: ATO-TS sends the requested SPs, where the location data in one SP is longer than the length of this SP	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
24	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
25	ATO-TS: ATO-TS sends a JP Overwrite	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
26	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
27	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
28	ATO-TS: ATO-TS sends the requested SPs, where one length data in one SP is zero	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
29	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
30	ATO-TS: ATO-TS sends a JP Overwrite	JP (overwrite status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
31	ATO-OB: ATO-OB sends a JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
32	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
33	ATO-TS: ATO-TS sends the requested SPs, where the version of one SP received is not the one included in the corresponding JP	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
34	ATO-OB: ATO-OB detects JP-SP inconsistency	STR issued by ATO-OB and received by the ATO-TS mentioning a data inconsistency.	Pass / Fail	
35	ATO-TS: ATO-TS receives an STR indicating that JP-SP inconsistency has been detected (condition to end Test Case)	SP(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
Notes				

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7.6 ATO inhibition

ID	Date	Location / Line
6	<YYYY-MM-DD>	<Line>
Title		
ATO inhibition		
Description		
This describes the operational activities required when the train crossing an ATO Inhibition Zone.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP including an "ATO inhibition zone" located in advance of the Train	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB enters the "ATO Inhibition zone"			
4	ATO-OB: ATO-OB detects that the Operational Conditions are lost			
5	ATO-OB: ATO-OB transitions to DE State	STR issued by ATO-OB and received by the ATO-TS mentioning that the ATO-OB is in DE State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	Operational Staff: Before 5 seconds have passed since entering DE State, the driver moves the TBL to brake position			
7	ATO-OB: ATO-OB detects that the TBL is in brake position			
8	ATO-OB: ATO-OB transitions to NA State	STR issued by ATO-OB and received by the ATO-TS mentioning that the ATO-OB is in NA State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	External System: ETCS-OB transitions to FS Mode	"FS Mode" displayed on ETCS-OB DMI.	Pass / Fail	
10	Operational Staff: The driver moves the TBL to traction position			
11	ATO-OB: ATO-OB leaves the "ATO Inhibition zone"			
12	ATO-OB: ATO-OB transitions to RE State			
13	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in RE State (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that the ATO-OB is in RE State. STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		It has been decided to not simulate the situation where the driver does not acknowledge the ATO disengagement, as the consequences are purely train related		

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7.7 C-DAS inhibition

ID	Date	Location / Line
7	<YYYY-MM-DD>	<Line>
Title		
C-DAS inhibition		
Description		
This describes the operational activities required when the train is crossing a DAS Inhibition Zone.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP including an “DAS inhibition zone” located in advance of the Train	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB enters the “DAS Inhibition zone”			
4	ATO-OB: ATO-OB will stop requesting ETCS-OB to display “Stopping Points distance”, “Stopping Point name and estimated arrival time”, and “DAS information”	“Stopping Points distance”, “Stopping Point name and estimated arrival time”, and “DAS information” no more displayed on ETCS-OB DMI	Pass / Fail	
5	ATO-OB: ATO-OB leaves the “DAS Inhibition zone”		Pass / Fail	
6	ATO-OB: ATO-OB will request ETCS-OB to display “Stopping Points distance”, “Stopping Point name and estimated arrival time”, and “DAS information” (condition to end Test Case)	“Stopping Points distance”, “Stopping Point name and estimated arrival time”, and “DAS information” are displayed on ETCS-OB DMI	Pass / Fail	
Notes		-		

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7.8 Train stops at a Stopping Point

ID	Date	Location / Line
8	<YYYY-MM-DD>	<Line>
Title		
Train stops at a Stopping Point		
Description		
This describes the operational activities required for an ATO train to stop at a stopping point.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with two additional Timing Points, the first being a Stopping Point and the second being a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB comes to stop at the Stopping Point and the Stopping Point is considered as reached according to the Q_STP_Reached of the SP's respective timing point."	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-OB: ATO-OB detects accurate stopping (i.e. within the stopping window)	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	ATO-OB: ATO-OB request the ETCS-OB to display that accurate stop has been achieved	"Inside Stopping Window" displayed on ETCS-OB DMI.	Pass / Fail	
8	ATO-OB: ATO-OB transitions to AV State	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	ATO-OB: ATO-OB will request the ETCS-OB to display the remaining dwell time	Remaining dwell time" displayed on ETCS-OB DMI	Pass / Fail	
10	ATO-OB: ATO-OB will request the ETCS-OB to display which sides the doors should be opened	"Door opening request" displayed on the DMI	Pass / Fail	
11	Operational Staff: The driver opens the doors (condition to end Test Case)	"Doors are open" displayed on the DMI	Pass / Fail	
Notes		- The opening/release of the doors when the train has stopped accurately can be configured to be automatic or manual.		

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7.9 Train stopped by signalling

ID	Date	Location / Line
9	<YYYY-MM-DD>	<Line>
Title		
Train stopped by signalling		
Description		
This describes the operational activities when a train is automatically driven and stopped by signalling. The train shall not disengage but continue when granted a new movement authority.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB comes to a stop in rear of the EOA	STR issued by ATO-OB and received by the ATO-TS mentioning that the train is stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
2	External Systems: ETCS-OB receives a new MA and sends speed and distance of new MA (EOA) to ATO-OB. (V_EOALOA, N_LOC_EOALOA)			
3	ATO-OB: ATO-OB updates the Operational Speed Profile			
4	ATO-OB: ATO-OB starts to request traction and the train starts to move			
5	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is moving (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		- When a train is stopped by signalling and driving automatically in GoA2, it will drive until the EOA and stop, but will not disengage. When a movement authority is received by on-board, the ATO-OB will continue to drive automatically without any further command required from the driver		

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7.10 Train stops short at a Stopping Point

ID	Date	Location / Line
10	<YYYY-MM-DD>	<Line>
Title		
Train stops short at a Stopping Point		
Description		
This describes the operational activities required for an ATO train to jog forward after stopping short of a stopping point.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with two additional Timing Points, the first being a Stopping Point and the second being a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB comes to a stop at the Stopping Point and the Stopping Point is considered as reached according to the Q_STP_Reached of the SP's respective timing point."	STR issued by ATO-OB and received by the ATO-TS mentioning that the train is stopped and mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-OB: ATO-OB detects that the train has undershot the stopping window			
6	ATO-OB: ATO-OB request the ETCS-OB to display that the train has undershot the stopping window	"Undershot" displayed on ETCS-OB DMI And STR issued by ATO-OB and received by the ATO-TS mentioning "undershoot" And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
7	ATO-OB: ATO-OB will maintain the train stationary by requesting the application of the Train Holding Brake			
8	ATO-OB: ATO-OB transitions to AV State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in AV State and mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	Operational Staff: The driver begin to move the train to manually correct the train position			
10	ATO-OB: ATO-OB will stop requesting the ETCS-OB to display that the train has undershot the stopping window	"Undershot" removed from ETCS-OB DMI	Pass / Fail	
11	Operational Staff: The driver stops the train within the stopping window of the Stopping Point			
12	ATO-OB: ATO-OB detects accurate stopping (i.e. within the stopping window)			
13	ATO-OB: ATO-OB request the ETCS-OB to display that accurate stop has been achieved	"Accurate stop" displayed on ETCS-OB DMI And	Pass / Fail	

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		STR issued by ATO-OB and received by the ATO-TS mentioning “accurate stop” And STRAck issued by ATO-TS and received by the ATO-OB.		
14	ATO-OB: ATO-OB will request the ETCS-OB to display the remaining dwell time	“Remaining dwell time” displayed on ETCS-OB DMI	Pass / Fail	
15	ATO-OB: ATO-OB will request the ETCS-OB to display which sides the doors should be opened	“Door opening request” displayed on the DMI	Pass / Fail	
16	Operational Staff: The driver opens the doors (condition to end Test Case)		Pass / Fail	
Notes		- The opening/release of the doors when the train has stopped accurately can be configured to be automatic or manual.		

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7.11 Train overshoots at Stopping Point

ID	Date	Location / Line
11	<YYYY-MM-DD>	<Line>
Title		
Train overshoots at Stopping Point		
Description		
This describes the operational activities required when an ATO train overshoots a stopping point.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with two additional Timing Points, the first being a Stopping Point and the second being a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB comes to a stop at the Stopping Point and the Stopping Point is considered as reached according to the Q_STP_Reached of the SP's respective timing point."	STR issued by ATO-OB and received by the ATO-TS mentioning that the train is stopped and mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-OB: ATO-OB detects that the train has overshot the stopping window			
6	ATO-OB: ATO-OB request the ETCS-OB to display that the train has overshot the stopping window	"Overshoot" displayed on ETCS-OB DMI And STR issued by ATO-OB and received by the ATO-TS mentioning "overshoot" And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
7	ATO-OB: ATO-OB will maintain the train stationary by requesting the application of the Train Holding Brake			
8	ATO-OB: ATO-OB transitions to AV State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in AV State and mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	Operational Staff: The driver begin to move the train to manually correct the train position			
10	ATO-OB: ATO-OB will stop requesting the ETCS-OB to display that the train has overshot the stopping window	"Overshoot" removed from ETCS-OB DMI	Pass / Fail	
11	Operational Staff: The driver stops the train within the stopping window of the Stopping Point			
12	ATO-OB: ATO-OB detects accurate stopping (i.e. within the stopping window)			
13	ATO-OB: ATO-OB request the ETCS-OB to display that accurate stop has been achieved	"Accurate stop" displayed on ETCS-OB DMI And	Pass / Fail	

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		STR issued by ATO-OB and received by the ATO-TS mentioning “accurate stop” And STRAck issued by ATO-TS and received by the ATO-OB.		
14	ATO-OB: ATO-OB will request the ETCS-OB to display the remaining dwell time	“Remaining dwell time” displayed on ETCS-OB DMI	Pass / Fail	
15	ATO-OB: ATO-OB will request the ETCS-OB to display which sides the doors should be opened	“Door opening request” displayed on the DMI	Pass / Fail	
16	Operational Staff: The driver opens the doors (condition to end Test Case)			
Notes		- The opening/release of the doors when the train has stopped accurately can be configured to be automatic or manual. - If the ATO-OB overshoots the stopping point and brings the train to a stand beyond the rollback tolerance, then the driver will be informed and have to take manual action based on operational rules.		

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7.12 Train departs from a Stopping Point

ID	Date	Location / Line
12	<YYYY-MM-DD>	<Line>
Title		
Train departs from a Stopping Point		
Description		
This describes the operational activities required for an ATO train to depart from a station.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB detects that the Dwell time is equal to the time to close the doors			
2	ATO-OB: ATO-OB will request the ETCS-OB to display “Request driver to close doors”	“Door closing request” displayed on the DMI	Pass / Fail	
3	Operational Staff: The driver closes the doors			
4	ATO-OB: ATO-OB will request the ETCS-OB to display “Doors are closed”	“Door closed” displayed on the DMI	Pass / Fail	
5	ATO-OB: ATO-OB detects that the dwell time elapses			
6	ATO-OB: ATO-OB detects that all Engagement Conditions are fulfilled			
7	ATO-OB: ATO-OB transitions to RE State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in RE State and mentioning the ID of the Stopping Point the train is currently stopped. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
8	Operational Staff: The driver selects “ATO Engage”			
9	External Systems: ETCS-OB will report to ATO-OB that “ATO Engage” has been selected (by increasing the related counter in packet 7 in SUBSET-130)			
10	ATO-OB: ATO-OB detects that “ATO Engage” has been selected			
11	ATO-OB: ATO-OB transitions to EG State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in EG State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
12	ATO-OB: ATO-OB starts to request traction and the train starts to move			
13	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is moving (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the Stopping Point the train is currently departed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes				

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7.13 Rerouting

ID	Date	Location / Line
13	<YYYY-MM-DD>	<Line>
Title		
Rerouting		
Description		
This describes the operational activities when a train is receiving an updated JP including a Stopping Point which is too close to permit the train to stop. In a second stage, this Test Case creates a "routing error" to be detected and managed by the ATO-OB.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-TS: ATO-TS sends a JP Update, starting at the SP where the train is located, with two Timing Points, the first being a Stopping Point (too close to the train to enable stopping) and the second being a Passing Point	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass/Fail	
2	ATO-OB: ATO-OB sends JPAck	JP Ack issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
3	ATO-OB: ATO-OB updates the Operational Speed Profile			
4	ATO-OB: ATO-OB detects that it is unable to stop at the next Stopping Point			
5	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is unable to stop at the next Stopping Point (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning "unable to stop at the next Stopping Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	ETCS-OB: ETCS-OB sends an MA including a linked balise which is not consistent with the current JP.			
7	ATO-OB: ATO-OB detects a "Routing Error".	STR issued by ATO-OB and received by the ATO-TS mentioning a routing error. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass/Fail	
8	ATO-TS: ATO-TS sends a JP Update, starting at the SP where the train is located, with one Timing Points being a Passing Point	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass/Fail	
Notes		-		

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7.14 Trackside initiated “Stopping Point Skip”

ID	Date	Location / Line
14	<YYYY-MM-DD>	<Line>
Title		
Trackside initiated “Stopping Point Skip”		
Description		
This describes the operational activities required for an ATO train to skip a stopping point when commanded by the trackside.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with two additional Timing Points, the first being a Stopping Point and the second being a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-TS: ATO-TS sends a JP Update where the next Stopping Point is marked to be skipped	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-OB: ATO-OB sends JPAck	JP Ack issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
6	ATO-OB: ATO-OB detects that the next Stopping Point is to be skipped			
7	ATO-OB: ATO-OB will request ETCS-OB to display “Skip Stopping Point requested by the ATO-TS”	“Skip Stopping Point requested by the ATO-TS” displayed on ETCS-OB DMI	Pass / Fail	
8	ATO-OB: ATO-OB updates the Operational Speed Profile			
9	ATO-OB: ATO-OB passes the Stopping Point to be skipped			
10	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB has passed the Stopping Point to be skipped (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that the Stopping Point to be skipped has been passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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7.15 Driver initiated “Stopping Point Skip”

ID	Date	Location / Line
15	<YYYY-MM-DD>	<Line>
Title		
Driver initiated “Stopping Point Skip”		
Description		
This describes the operational activities required for an ATO train to skip a stopping point when commanded by the driver.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with two additional Timing Points, the first being a Stopping Point and the second being a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	Operational Staff: The driver selects “Skip Stopping Point Request”			
5	External Systems: ETCS-OB will report to ATO-OB that “Skip Stopping Point Request” has been selected (by increasing the related counter in packet 7 in SUBSET-130)			
6	ATO-OB: ATO-OB detects that “Skip Stopping Point Request” has been selected			
7	ATO-OB: ATO-OB will request ETCS-OB to display “Skip Stopping Point requested by the driver”	“Skip Stopping Point requested by the driver” displayed on ETCS-OB DMI	Pass / Fail	
8	ATO-OB: ATO-OB updates the Operational Speed Profile			
9	ATO-OB: ATO-OB passes the Stopping Point to be skipped			
10	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB has passed the Stopping Point to be skipped (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that the Stopping Point to be skipped has been passed. And STRAck issued by ATO-TS and received by the ATO-OB.		
Notes		-		

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7.16 Hold train at a Stopping Point

ID	Date	Location / Line
16	<YYYY-MM-DD>	<Line>
Title		
Hold train at a Stopping Point		
Description		
This describes the operational activities required to apply a train hold command.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	ATO-TS: Before the dwell time elapses, the ATO-TS sends a JP Update where Train Hold at Stopping Point is requested	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
2.	ATO-OB: ATO-OB sends JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
3.	ATO-OB: ATO-OB will set the dwell time to hold			
4.	ATO-OB: ATO-OB will request ETCS-OB to display “Train Hold”	“Train Hold” displayed on ETCS-OB DMI	Pass / Fail	
5.	ATO-TS: ATO-TS sends a JP Update with a new departure time	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6.	ATO-OB: ATO-OB sends JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
7.	ATO-OB: ATO-OB will stop requesting ETCS-OB to display “Train Hold” (condition 1 to end Test Case)	“Train Hold” removed from ETCS-OB DMI.	Pass / Fail	
8.	ATO-OB: ATO-OB will request the ETCS-OB to display the remaining dwell time (condition 2 to end Test Case)	“Remaining dwell time” displayed on ETCS-OB DMI.	Pass / Fail	
9.	ATO-TS: ATO-TS receives the JPAck (condition 3 to end Test Case)			
Notes		- The train is held until a new Journey Profile is received or driver manually departs.		

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7.17 ATO-OB reaching AV State

ID	Date	Location / Line
17	<YYYY-MM-DD>	<Line>
Title		
ATO-OB reaching AV State		
Description		
This describes the operational activities required to acquire and manage the relevant information to reach the AV state		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS with an undefined SP as a reference	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with only one Timing Point, which is a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: Upon reception of the JP, the ATO-OB will start to monitor the triggering conditions for sending STRs			
4	ATO-OB: ATO-OB detects that all Operational Conditions are fulfilled			
5	ATO-OB: ATO-OB transitions to AV State			
6	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in AV State (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in AV State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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7.18 ATO-OB reaching RE State

ID	Date	Location / Line
18	<YYYY-MM-DD>	<Line>
Title		
ATO-OB reaching RE State		
Description		
This describes the operational activities required to reach the RE state.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	Manual operator: Driver moves the Direction controller to forward position			
2	Manual operator: Driver moves the TBL to traction position and starts to move the train			
3	ATO-OB: ATO-OB detects that all Engagement Conditions are fulfilled			
4	ATO-OB: ATO-OB transitions to RE State	STR issued by ATO-OB and received by the ATO-TS mentioning that ATO-OB is in RE State. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes				

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7.19 Handover between two adjacent ATO-TS

ID	Date	Location / Line
19	<YYYY-MM-DD>	<Line>
Title		
Handover between two adjacent ATO-TS		
Description		
This describes the operational activities required to perform a hand-over between two adjacent ATO-TS. In a first stage, the train is foreseen to go in the direction of a given Accepting ATO-TS. And, in a second stage, the train is rerouted to go to another Accepting ATO-TS.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS evaluates that the available storage capacity does not allow to include the SP that contains ATO-TS Contact Information and sends a JP including only the SPs in rear of the one containing ATO-TS Contact Information	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes a Passing Point and makes additional storage capacity available	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
5	ATO-TS: ATO-TS evaluates that the available storage capacity allows to include the SP that contains ATO-TS Contact Information and sends a JP including the SP containing ATO-TS Contact Information	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	ATO-OB: When the train approaches the border of the area controlled by the current ATO-TS (“Handing Over ATO-TS”), the ATO-OB sends an HSReq to the adjacent ATO-TS (“Accepting ATO-TS”), using the Contact Information provided in the SP, indicating that it is due to a handing over procedure	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
7	ATO-TS: The Accepting ATO-TS sends HSAck	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
8	ATO-TS: The Handing Over ATO-TS sends a JP Update where the last SP contains deviating ATO-TS Contact Information to an adjacent ATO-TS.	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	ATO-OB: ATO-OB sends JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
10	ATO-OB: ATO-OB sends the packet SESSTerm to the Accepting ATO-TS and terminates the communication session	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
11	ATO-OB: When the train approaches the border of the area controlled by the current ATO-TS (“Handing Over ATO-TS”), the ATO-OB sends an HSReq to the adjacent ATO-TS (“Accepting ATO-TS”), using the Contact Information provided in the SP, indicating that it is due to a handing over procedure	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
12	ATO-TS: The Accepting ATO-TS sends HSAck with cyclic reporting configured to be 1 second	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
13	ATO-OB: ATO-OB sends a JPReq to the Accepting ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
14	ATO-TS: Accepting ATO-TS sends a valid maximum size JP including the maximum number of referenced maximum size SPs, where the first Timing Point is a Passing point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	

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15	ATO-OB: ATO-OB process the data in the maximum JP and corresponding SPs			
16	ATO-OB: Upon reception of the JP, the ATO-OB will start to monitor the triggering conditions for sending STRs to the Accepting ATO-TS			
17	ATO-OB: ATO-OB detects that the reporting time interval for triggering an STR to the Accepting ATO-TS is elapsed	STR issued by ATO-OB and received by the Accepting ATO-TS mentioning the ID of the last Passing Point passed	Pass / Fail	
18	ATO-OB: ATO-OB sends a JPReq to the Accepting ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
19	ATO-TS: Accepting ATO-TS sends a JP without any additional information, as the train is not completely within its ATO-TS area and the Accepting ATO-TS has already sent one maximum JP including the maximum number of referenced maximum size SPs	JP (unavailable status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
20	ATO-OB: ATO-OB passes the last Passing Point included in the JP which ends at the boundary of the ATO-TS (received from the Handing Over ATO-TS)	STR issued by ATO-OB and received by both the Handing Over ATO-TS and the Accepting ATO-TS mentioning the ID of the last Passing Point passed.	Pass / Fail	
21	ATO-TS: Handing Over ATO-TS and Accepting ATO-TS sends STRAck	STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
22	ATO-OB: ATO-OB will stop to monitor the triggering conditions for sending STRs to the Handing Over ATO-TS			
23	ATO-OB: ATO-OB detects that the rear of the train has left the last SP referred to by the JP which ends at the boundary of the ATO-TS (received from the Handing Over ATO-TS)			
24	ATO-OB: ATO-OB detects that Communication session transition condition [13] is fulfilled and sends the packet SESSTerm to the Handing Over ATO-TS and terminates the communication session	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
25	ATO-OB: ATO-OB regularly sends a STR giving the current train position and length in accordance with “T_Reporting_Time”.	STR issued by ATO-OB and received by the ATO-TS giving the train position and length. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
26	ATO-TS: ATO-TS detects that the train is completely inside the Accepting ATO-TS area			
27	ATO-TS: ATO-TS sends a JP Update including the maximum number of referenced maximum size SPs	JP (update status) issued by Accepting ATO-TS and received by the ATO-OB.	Pass / Fail	
28	ATO-OB: ATO-OB stores both received maximum size JPs including the maximum number of referenced maximum size SPs			
29	ATO-TS: ATO-TS sends a JP Update removing all Timing Points in the previous JP except the first one, which is a Passing Point	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
30	ATO-OB: ATO-OB sends JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
31	ATO-TS: ATO-TS receives the JPAck (condition to end Test Case)			
Notes		-		

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7.20 ATO Adhesion Management

ID	Date	Location / Line
20	<YYYY-MM-DD>	<Line>
Title		
ATO Adhesion Management		
Description		
This describes the operational activities required for ATO trains to report requirements to change adhesion settings. This Test Case includes the different way acquiring low adhesion information: slippery rail information given by the driver to ETCS-OB, low adhesion area given via a JP and slip/slide information given by the Train.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	Operational Staff: Driver selects “low adhesion” on ETCS-OB DMI (as an ETCS function).			
2	External System: ETCS-OB reports the information “slippery rail” to the ATO-OB			
3	ATO-OB: ATO-OB detects that low adhesion is selected by the driver and the STR is issued due to Q_STR_Indicators low adhesion reported by the driver.	STR issued by ATO-OB and received by the ATO-TS mentioning that Low Adhesion is reported by the driver. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-TS: ATO-TS sends a JP Update including a “Low adhesion Area” located in advance of the Train	JP (update status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	ATO-OB: ATO-OB sends JPAck	JPAck issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
7	ATO-OB: ATO-OB updates the Operational Speed Profile		Pass / Fail	
8	External System: The train reports “slip/slide” information		Pass / Fail	
9	ATO-OB: ATO-OB detects that the train reports “slip/slide” information			
10	ATO-TS: ATO-TS receives an STR indicating that Slip/Slide information has been detected by external system and the STR is issued due to Q_STR_Indicators “Slip/slide reported by TCMS/Train”	STR issued by ATO-OB and received by the ATO-TS mentioning that slip/slide has been reported by the Train. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes				

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7.21 Change of Train Running Number

ID	Date	Location / Line
21	<YYYY-MM-DD>	<Line>
Title		
Change of Train Running Number		
Description		
This describes the operational activities required when a Train Running Number is changed.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	Operational Staff: Driver changes the TRN			
2.	ATO-OB: ATO-OB detects that the TRN has changed			
3.	ATO-OB: ATO-OB detects that Communication session transition condition [4] is fulfilled and sends the packet SESSTerm to the ATO-TS and terminates the communication session	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
4.	ATO-OB: ATO-OB deletes the stored JP			
5.	ATO-OB: ATO-OB detects that the Operational Conditions are lost			
6.	ATO-OB: ATO-OB transitions to DE State			
7.	ATO-OB: ATO-OB sends an HSReq to ATO-TS using the pre-configured ATO-TS Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
8.	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval			
9.	ATO-OB: ATO-OB receives the HSAck (condition 1 to end Test Case)	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
10.	Operational Staff: Before 5 seconds have passed since entering DE State, the driver moves the TBL to brake position			
11.	ATO-OB: ATO-OB detects that the TBL is in brake position			
12.	ATO-OB: ATO-OB transitions to NA State (condition 2 to end Test Case)			
13.	ATO-OB: ATO-OB detects that the train has come to a stop (condition 3 to end Test Case)			
Notes		-		

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7.22 Transition from Class B area

ID	Date	Location / Line
22	<YYYY-MM-DD>	<Line>
Title		
Transition from Class B area		
Description		
This describes the operational activities required when a Train is leaving a Class B system (Level NTC) and transitioning to ETCS		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
4	ATO-TS: ATO-TS sends a JP Update with only one Timing Point, which is a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-OB: Upon reception of the JP, the ATO-OB will start to monitor the triggering conditions for sending STRs			
6	External System: The train enters the ETCS area and ETCS-OB transitions to FS Mode			
7	ATO-OB: ATO-OB detects that all Operational Conditions are fulfilled			
8	ATO-OB: ATO-OB transitions to AV State			
9	ATO-OB: ATO-OB detects that all Engagement Conditions are fulfilled			
10	ATO-OB: ATO-OB transitions to RE State			
11	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in RE State (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes				

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7.23 Transition to Class B area

ID	Date	Location / Line
23	<YYYY-MM-DD>	<Line>

Title

Transition to Class B area

Description

This describes the operational activities required when a Train is leaving an ETCS area and transitioning to a Class B system area (Level NTC). It is assumed that the last Timing Point is in the ETCS area. When passing the last passing point, the ATO-OB transitions to NA while the ETCS-OB transitions to FS.

Test Cases

Sequences of the Test Case

Step	Step description	Description of what to be tested	Statement	Comment
1.	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested.	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2.	ATO-TS: ATO-TS sends a valid JP with only one additional Timing Point which is a Passing Point located in front of the train just before the border of a Class B area.	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3.	ATO-OB: ATO-OB passes the first Passing Point.	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4.	ATO-OB: ATO-OB passes the last Passing Point included in the JP	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5.	ATO-OB: ATO-OB will stop to monitor the triggering conditions for sending STRs			
6.	ATO-OB: ATO-OB detects that the Operational Conditions are lost			
7.	ATO-OB: ATO-OB transitions to DE State			
8.	Operational Staff: Before 5 seconds have passed since entering DE State, the driver moves the TBL to brake position			
9.	ATO-OB: ATO-OB detects that the TBL is in brake position			
10.	ATO-OB: ATO-OB transitions to NA State			
11.	External System: ETCS-OB leaves AD Mode.	“FS Mode” displayed on the DMI.	Pass / Fail	
12.	Operational Staff: The driver moves the TBL to traction position			
13.	External System: The train enters the Class B area.		Pass / Fail	
14.	ATO-OB: ATO-OB detects that the rear of the train has left the last SP referred to by the JP			
15.	ATO-OB: ATO-OB detects that Communication session transition condition [13] is fulfilled and sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
Notes				

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7.24 Invalid packet received by ATO-OB

ID	Date	Location / Line
24	<YYYY-MM-DD>	<Line>
Title		
Invalid packet received by ATO-OB		
Description		
This describes the operational activities related to the reception of invalid packets. It must be noted that this Test Case can be performed with a simulated ATO-TS only. This Test Case is based on the sending of JP Update. The time between the sending of each JP Update shall be greater than the current cycle time considered for the sending of STR.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-TS: ATO-TS sends a JP Update where the packet number is not among the list of defined packet numbers	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
2	ATO-OB: ATO-OB detects that the received packet has an unknown packet number (packet will be ignored)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-TS: ATO-TS sends a JP Update with a timestamp that is older than the timestamp of the previously received JP packet	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
4	ATO-OB: ATO-OB detects that the received packet has a timestamp that is older than the timestamp of a previously received packet with the same number (packet will be ignored)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
5	ATO-TS: ATO-TS sends a JP Update with a packet counter that has not changed compared to the last JP packet	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
6	ATO-OB: ATO-OB detects that the packet counter of the received packet has not changed compared to the last received packet with the same number (packet will be ignored)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
7	ATO-TS: ATO-TS sends a JP Update where one of the variables uses a spare value	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
8	ATO-OB: ATO-OB detects that the one of the variables in the received packet uses a spare value (packet will be ignored)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
9	ATO-TS: ATO-TS sends a JP Update where the Header refers to a TRN that does not match the TRN provided by the ETCS-OB	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
10	ATO-OB: ATO-OB detects that the Header of the received packet refers to a TRN that does not match the TRN provided by the ETCS-OB (packet will be ignored)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	

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11	ATO-TS: ATO-TS sends a JP Update where the Header refers to an ETCS Identity that does not match the ETCS Identity provided by the ETCS-OB	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
12	ATO-OB: ATO-OB detects that the Header of the received packet refers to an ETCS Identity that does not match the ETCS Identity provided by the ETCS-OB (packet will be ignored) (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS with no update of the ID of the next targeted Passing Point. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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7.25 Loss of communication Session

ID	Date	Location / Line
25	<YYYY-MM-DD>	<Line>
Title		
Loss of communication Session		
Description		
This describes the operational activities related to the loss of a Communication Session including re-establishment with the ATO-TS corresponding to the preconfigured Contact Information. It must be noted that this Test Case can be performed with a simulated ATO-TS only.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2.	ATO-OB: ATO-OB detects that no JP has been received before the end of the "Timeout for ATO-TS response" mentioned in the HSAck		Pass / Fail	
3.	ATO-OB: ATO-OB sends a HSReq to the ATO-TS using the same Contact Information that was used when establishing the Session with the ATO-TS	HSReq issued by ATO-OB and received by the ATO-TS	Pass / Fail	
4.	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
5.	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
6.	ATO-TS: ATO-TS sends a valid JP containing two additional Passing Points; the first one located close after the existing Passing Point.	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
7.	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
8.	ATO-OB: ATO-OB detects that no SP has been received before the end of the "Timeout for ATO-TS response" mentioned in the HSAck			
9.	ATO-OB: ATO-OB sends a HSReq to the ATO-TS using the same Contact Information that was used when establishing the session with the ATO-TS	HSReq issued by ATO-OB and received by the ATO-TS	Pass / Fail	
10.	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
11.	ATO-OB: ATO-OB sends SPReqs to ATO-TS corresponding to all the SPs identified in the received JP	SPReq(s) issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
12.	ATO-TS: ATO-TS sends the requested SPs	SP(s) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
13.	ATO-OB: ATO-OB passes a Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
14.	ATO-TS: ATO-TS sends an STRAck with a timestamp and packet counter information that are not equal to the values of the packet to be acknowledged			
15.	ATO-OB: ATO-OB detects that the timestamp and packet counter information of the received STRAck packet are not equal to the values of the packet to be acknowledged			

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16.	ATO-OB: ATO-OB passes a Passing Point before the end of the "Timeout for ATO-TS" response.	No STR is sent by the ATO-OB because the previous STR has not yet been acknowledged by the ATO-TS.	Pass / Fail	
17.	ATO-OB: ATO-OB detects that no (valid) STRAck packet has been received before the end of the "Timeout for ATO-TS response" mentioned in the HSAck			
18.	ATO-OB: ATO-OB sends a HSReq to the ATO-TS using the same Contact Information that was used when establishing the Session with the ATO-TS	HSReq issued by ATO-OB and received by the ATO-TS	Pass / Fail	
19.	ATO-OB: ATO-OB passes the last Passing Point included in the JP without having received an HSAck			
20.	ATO-OB: ATO-OB detects that the Operational Conditions are lost			
21.	ATO-OB: ATO-OB transitions to DE State			
22.	Operational Staff: Before 5 seconds have passed since entering DE State, the driver moves the TBL to brake position			
23.	ATO-OB: ATO-OB detects that the TBL is in brake position			
24.	ATO-OB: ATO-OB transitions to NA State			
25.	Operational Staff: The driver moves the TBL to traction position and the train is moving according to a new route back towards the area covered by the preconfigured ATO-TS.			
26.	ATO-OB: ATO-OB sends an HSReq to ATO-TS using the pre-configured ATO-TS Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
27.	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
28.	ATO-OB: ATO-OB receives the HSAck			
29.	ATO-OB: ATO-OB sends a JPReq to ATO-TS with an undefined SP as a reference	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
30.	ATO-TS: ATO-TS sends a valid JP with only one Timing Point, which is a Passing Point	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
31.	ATO-OB: Upon reception of the JP, the ATO-OB will start to monitor the triggering conditions for sending STRs			
32.	Operational Staff: The train crosses the border to the area controlled by the preconfigured ATO-TS			
33.	ATO-OB: ATO-OB detects that all Operational Conditions are fulfilled			
34.	ATO-OB: ATO-OB transitions to AV State	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
35.	ATO-OB: ATO-OB detects that all Engagement Conditions are fulfilled			
36.	ATO-OB: ATO-OB transitions to RE State			
37.	ATO-TS: ATO-TS receives an STR indicating that the ATO-OB is in RE State (condition to end Test Case)	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
38.				
Notes				

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7.26 Communication session termination (SESSTermReq sent by ATO-TS)

ID	Date	Location / Line
26	<YYYY-MM-DD>	<Line>
Title		
Communication session termination (SESSTermReq sent by ATO-TS)		
Description		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-TS: ATO-TS sends SESSTermReq			
2	ATO-OB: ATO-OB sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
Notes		-		

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7.27 Communication session termination (Cab is closed)

ID	Date	Location / Line
27	<YYYY-MM-DD>	<Line>
Title		
Communication session termination (Cab is closed)		
Description		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	Operational Staff: The driver closes the cab			
2.	ATO-OB: ATO-OB sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition 1 to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
3.	Operational Staff: The driver opens the cab and ETCS-OB reaches FS Mode (condition 2 to end Test Case)			
Notes		-		

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7.28 Communication session termination (invalid TRN or train length)

ID	Date	Location / Line
28	<YYYY-MM-DD>	<Line>
Title		
Communication session termination (invalid TRN or train length)		
Description		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	External System: ETCS-OB detects invalid TRN and sends the information “ETCS Operational Data not valid” to the ATO-OB and stops sending the TRN to the ATO-OB			
2.	ATO-OB: ATO-OB sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
Notes		-		

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7.29 Communication session termination (ETCS-OB is in NL Mode or in SH mode)

ID	Date	Location / Line
29	<YYYY-MM-DD>	<Line>
Title		
Communication session termination (ETCS-OB is in NL Mode or in SH mode)		
Description		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1.	External System: ETCS-OB transitions to NL Mode			
2.	ATO-OB: ATO-OB sends the packet SESSTerm to the ATO-TS and terminates the communication session (condition 1 to end Test Case)	Packet SESSTerm issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
3.	External System: ETCS-OB transitions from NL Mode (not to SH Mode) (condition 2 to end Test Case)			
Notes		-		

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7.30 Communication session establishment (Degraded Modes)

ID	Date	Location / Line
33	<YYYY-MM-DD>	<Line>
Title		
Communication session establishment (Degraded Modes)		
Description		
This describes the operational activities required to establish a communication session when in NA State, either after energising the ATO onboard or after a change of TRN. The following use cases are considered in this Test Case: • An HSReq and no feedback is received before 30 seconds; • An HSRej is sent giving a Contact Information; • An HSRej is sent indicating that the ATO-TS and ATO-OB are not compatible; • An HSRej is sent indicating that the ATO-TS in charge is not known; • A valid HSAck is sent.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB detects that ETCS-OB sends validated Train Data			
2	ATO-OB: ATO-OB sends an HSReq to ATO-TS using the pre-configured ATO-TS Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
3	ATO-OB: ATO-OB detects that no feedback has been received after 30 seconds, and sends a new HSReq, using the same ATO-TS Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS after 30 seconds.	Pass / Fail	
4	ATO-TS: ATO-TS sends HSRej indicating that another ATO-TS is in charge, and provides the Contact Information to the ATO-TS in charge	HSRej issued by ATO-TS and received by the ATO-OB	Pass / Fail	
5	ATO-OB: ATO-OB sends a HSReq to the ATO-TS using the Contact Information received in the HSRej	HSReq issued by ATO-OB and received by the ATO-TS mentioned in the HSRej.	Pass / Fail	
6	ATO-TS: ATO-TS sends HSRej indicating that the ATO-OB and ATO-TS are not compatible	HSRej issued by ATO-TS and received by the ATO-OB	Pass / Fail	
7	Operational Staff: Driver changes TRN	TRN changed on ETCS-OB DMI	Pass / Fail	
8	External: Validated Train Data is sent from ETCS-OB.			
9	ATO-OB: ATO-OB detects that ETCS-OB sends validated Train Data			
10	ATO-OB: When the ATO-OB detects that the TRN has changed, the ATO-OB sends a HSReq to ATO-TS using the pre-configured Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
11	ATO-TS: ATO-TS sends HSRej indicating that the ATO-TS in charge is not known	HSRej issued by ATO-TS and received by the ATO-OB	Pass / Fail	
12	ATO-OB: ATO-OB sends a HSReq to ATO-TS using the pre-configured Contact Information	HSReq issued by ATO-OB and received by the Preconfigured ATO-TS	Pass / Fail	
13	ATO-TS: ATO-TS sends HSAck with cyclic reporting configured at the maximum time interval	HSAck issued by ATO-TS and received by the ATO-OB	Pass / Fail	
14	ATO-OB: Communication session established (condition to end Test Case)			
Notes		-		

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7.31 ATO-OB receives invalid JP

ID	Date	Location / Line
24	<YYYY-MM-DD>	<Line>
Title		
ATO-OB receives Invalid JP		
Description		
This describes the operational activities related to the reception of an invalid JP after a JPReq sent by the AO-OB.		
Test Cases		
Sequences of the Test Case		

Step	Step description	Description of what to be tested	Statement	Comment
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating a wrong SP which does not belong to in the current JP	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a JP with the status “invalid”.	JP (invalid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
1	ATO-OB: ATO-OB sends a JPReq to ATO-TS indicating the last SP in the current JP as the SP from where the new JP is requested	JPReq issued by ATO-OB and received by the ATO-TS.	Pass / Fail	
2	ATO-TS: ATO-TS sends a valid JP with one additional Timing Point which is a Passing Point.	JP (valid status) issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
3	ATO-OB: ATO-OB passes the first Passing Point	STR issued by ATO-OB and received by the ATO-TS mentioning the ID of the last Passing Point passed. And STRAck issued by ATO-TS and received by the ATO-OB.	Pass / Fail	
Notes		-		

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8. TRACEABILITY MATRIX

8.1.1.1 The traceability matrix gives:

- the requirements from the SUBSET-125 which are related to Track/Train communication;
- for each of them the main Test Case covering it & the list of Test Cases which are also related to.

8.1.1.2 Please refer to the separate file “SUBSET-151 - Traceability Matrix”.



SUBSET-151 - Traceability Matrix v

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